PROJECT REPORT

Streamlining IT Procurement: Automating Standard Laptop Orders with WI

Project Title	Streamlining IT Procurement: Automating standard laptop orders with WI
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This document describes an automated workflow system designed to simplify and speed up the process of ordering standard laptops in an organisation. The solution uses Workflow Intelligence (WI) to connect procurement requests, approvals, budget checks, vendor selection and inventory updates into one smooth digital process.

1. Abstract

Buying standard laptops for employees is a common but time-consuming task in many organisations. Usually the process involves paper forms or emails, long waiting times for manager approval, manual budget checks, and slow coordination with vendors and the inventory team. This project proposes an automated system that uses Workflow Intelligence (WI) to streamline the entire IT procurement cycle for standard laptop orders.

In the proposed system, an employee raises a procurement request through a simple digital form. The system then automatically routes the request for manager approval, performs a budget check, selects an approved vendor based on predefined rules, updates inventory records, and notifies all stakeholders. The main goal is to reduce delays, remove repetitive manual work, and make the laptop ordering process faster, more transparent and easier to track. This report explains the problem, objectives, existing manual process, the proposed automated solution, system architecture, step-by-step workflow, benefits, conclusion and future scope.

2. Introduction

Information Technology (IT) departments in companies regularly purchase laptops for new joiners, replacements and upgrades. When the number of requests is small, a manual process may work. But when many employees need standard laptops every month, the manual method creates bottlenecks. Emails get lost, approvals take days, budget verification is done offline, and the inventory team often does not know the status of pending orders.

Workflow Intelligence (WI) refers to the use of intelligent automation tools that can understand business rules, route work items, check conditions and trigger actions without constant human intervention. By applying WI to standard laptop procurement, organisations can convert a slow, email-driven process into a structured, trackable and mostly automatic workflow.

This project focuses only on standard laptop models (pre-approved configurations). Special or high-end custom machines are outside the current scope. The system is designed to be simple enough for final-year student implementation while still covering real business needs such as procurement request creation, manager approval, budget check, vendor selection and inventory update.

3. Problem Statement

The current way of ordering standard laptops in many organisations has several practical problems:

- **Manual request handling:** Employees send emails or fill paper forms. There is no single place to track the request.
- **Slow approval cycle:** Manager approval depends on email replies. If the manager is busy or on leave, the request waits for many days.
- **No automatic budget check:** Finance or IT has to manually verify whether enough budget is available for the department.
- **Vendor selection is inconsistent:** Different people may contact different vendors without following the approved vendor list.
- **Inventory is not updated on time:** After the laptop is ordered or delivered, stock records are often updated late or incorrectly.
- **Lack of visibility:** The requester, manager and IT team cannot easily see the current status of any order.

Because of these issues, laptop delivery is delayed, employees wait longer to start productive work, and the IT and procurement teams spend too much time on repetitive follow-ups instead of higher-value tasks.

4. Objectives

The main objectives of this project are:

1. To design a simple digital form for raising a standard laptop procurement request.
2. To automate the routing of the request to the reporting manager for approval.
3. To perform an automatic budget check based on department budget limits.
4. To select an approved vendor using predefined rules (price, delivery time, contract status).
5. To update the inventory system when an order is placed or a laptop is received.
6. To provide clear status visibility to the requester, manager and IT procurement team.
7. To reduce the overall time taken from request to order placement for standard laptops.

5. Existing System

In the existing system, the laptop ordering process is mostly manual and email-based. A typical flow looks like this:

1. Employee sends an email to the IT helpdesk or fills a paper request form stating the need for a laptop.
2. IT team checks if a standard configuration is required and forwards the request to the manager for approval.
3. Manager replies by email (approve / reject / ask for more information). This step often takes several days.
4. After approval, IT or procurement manually checks the remaining budget of the department.
5. If budget is available, someone contacts one or more vendors by email or phone to get quotations.
6. A purchase order is prepared and sent to the selected vendor.
7. When the laptop arrives, the inventory team updates the asset register, sometimes after a delay.
8. The employee is informed that the laptop is ready for collection or delivery.

This process depends heavily on individual follow-ups. There is no automatic reminder, no single dashboard, and no guaranteed order of steps. As a result, many requests get stuck between approval and budget check or between order placement and inventory update.

6. Proposed System

The proposed system replaces the email and paper-based process with a Workflow Intelligence (WI) driven automated workflow. All standard laptop orders follow the same structured path. Key features of the proposed system are:

- **Digital Procurement Request:** Employee fills a simple online form with employee ID, department, required laptop model (from a standard list), justification and required date.
- **Automatic Routing for Manager Approval:** The system identifies the reporting manager from the organisation hierarchy and sends an approval task with all necessary details.
- **Budget Check Module:** After approval, the system automatically checks the remaining budget allocated to that department for IT assets. If budget is insufficient, the request is flagged or sent back.

- **Vendor Selection Engine:** Based on rules (approved vendor list, best price for the standard model, delivery SLA), the system suggests or auto-selects the vendor.
- **Order Placement & Notification:** Once vendor is confirmed, a purchase order is generated and notifications are sent to procurement, vendor and requester.
- **Inventory Update:** When the laptop is marked as received, the inventory module is updated with asset tag, model, user assignment and status.
- **Status Dashboard:** Every stakeholder can see the current stage of the request (Pending Approval / Budget Check / Vendor Selected / Ordered / Delivered).

The system is designed so that most routine cases (standard laptop, approved budget, active vendor) can complete with minimal human intervention. Only exceptions (budget shortage, special approval, vendor issue) require extra attention.

7. System Architecture & Workflow

The overall architecture consists of five main modules connected through a central Workflow Intelligence engine:

1. Request Portal 2. Approval Engine 3. Budget Service 4. Vendor & Order Service 5. Inventory Service

All modules share a common request ID so that status can be tracked end-to-end. Below is the step-by-step laptop order automation workflow:

Step 1: Raise Procurement Request

Employee logs into the Request Portal and selects “Standard Laptop Order”. The form shows only approved laptop models. Employee fills required fields (name, department, model, reason, needed-by date) and submits. A unique Request ID is generated and the request enters the WI workflow.

Step 2: Manager Approval

WI engine looks up the reporting manager from the employee master data and creates an approval task. Manager receives a notification (email + dashboard). Manager can Approve, Reject or Request Clarification. If approved, the workflow moves to the next step. If rejected, the requester is informed and the process stops.

Step 3: Automatic Budget Check

The system calls the Budget Service with department code and estimated cost of the selected laptop model. If remaining budget ≥ required amount, the check passes and the workflow continues. If budget is low, the request is marked “Budget Hold” and a notification is sent to the department budget owner.

Step 4: Vendor Selection

Vendor & Order Service receives the approved and budget-cleared request. It applies selection rules: only vendors on the approved list, matching the laptop model, with valid contract, and preferably the lowest quote or best delivery time. The selected vendor and quoted price are attached to the request.

Step 5: Purchase Order Generation & Placement

A purchase order (PO) is automatically generated with Request ID, laptop model, quantity (usually 1), vendor details and delivery address. The PO is sent to the vendor through the integrated channel (email/API). Requester and procurement team receive confirmation.

Step 6: Inventory Update & Delivery Closure

When the laptop is received and asset-tagged, the Inventory Service is updated with serial number, asset ID, assigned user and location. The original request is marked “Completed”. The employee is notified that

the laptop is ready. All timestamps (request, approval, order, delivery) are stored for reporting.

Exception handling is built into each step. Timeouts, rejections and budget failures generate clear alerts so that no request remains stuck without visibility.

8. Benefits

Implementing the proposed WI-based automation brings the following benefits:

- **Faster turnaround:** Standard laptop orders that earlier took 1–2 weeks can be reduced to a few days because approvals and budget checks happen inside the workflow.
- **Less manual effort:** IT and procurement staff no longer need to chase emails or maintain separate spreadsheets for each request.
- **Consistent process:** Every standard laptop follows the same steps, reducing mistakes in vendor selection and budget usage.
- **Better budget control:** Automatic budget check prevents overspending and gives real-time visibility of remaining IT asset budget.
- **Improved inventory accuracy:** Asset records are updated as part of the same workflow, so the inventory always reflects current stock and assignments.
- **Transparency:** Requesters and managers can check status anytime instead of sending “any update?” emails.
- **Audit-ready records:** Every action (who approved, when, which vendor, what price) is logged against the Request ID.

9. Conclusion

This project shows that the common and repetitive task of ordering standard laptops can be made much more efficient by applying Workflow Intelligence. By connecting the procurement request, manager approval, budget check, vendor selection and inventory update into one automated flow, organisations can save time, reduce errors and give better service to employees who need a laptop to work.

The solution is deliberately kept simple and focused on standard configurations so that it is realistic to implement as a final-year project while still solving a real business problem. The same design principles can later be extended to other IT assets such as monitors, docking stations or mobile phones.

10. Future Scope

Possible future improvements and extensions include:

- Integration with the organisation’s HR system so that new joiner laptop requests are created automatically on the joining date.
- Mobile app or chatbot interface for raising and tracking requests.
- Machine learning based prediction of monthly laptop demand so that inventory can be pre-stocked.
- Support for non-standard / custom laptop configurations with extra approval levels.
- Direct API integration with major vendor portals for real-time quotation and order status.
- Dashboard with analytics: average approval time, budget utilisation by department, most ordered models, etc.
- Extension of the same WI framework to other procurement categories (software licences, peripherals, network equipment).

11. References

1. Microsoft Power Automate Documentation – Workflow automation concepts and best practices (learning.microsoft.com).

2. ITIL Foundation – Service Request Management and Asset Management practices.

3. Project Management Institute – Guidelines for process improvement and automation projects.

4. Academic papers and case studies on digital procurement transformation in mid-size organisations.

5. Internal organisational policies on IT asset procurement and approved vendor management (sample policy documents used for rule design).

6. Standard laptop configuration catalogues published by major hardware vendors (Dell, HP, Lenovo) for model selection rules.

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